The Food Group Student Data Challenge

The problem

The Food Group is part of the hunger-relief network in Minnesota. It works at two levels. A food bank is a warehouse that holds food and supplies other organizations. A food shelf is a site that gives food directly to households. The Food Group runs a food bank. It also serves food shelves and collects usage information.

The Food Group runs two further programs. It runs a call center that connects people to resources. It supports access to farmers markets through the MarketBucks program.

Each program creates data. The four core datasets are:

- Monthly visits, pounds distributed, and individuals served at each food shelf.
- Call logs with the reason for each call and the referral.
- Records of participation at farmers markets.
- Location and status records for each partner organization.

One question stays open. How much of this activity comes from food insecurity? How much of it comes from other causes? Food insecurity is the condition of a household that cannot get enough food for an active and healthy life.

A visit to a food shelf is one unit of demand. Demand is the help that people ask for. Need is the help that people must have. Demand and need are not the same.

Some visits come from households that go hungry without the food shelf. Other visits come from households with different reasons:

- To stretch a tight budget.
- To add to the food they already have.
- To use a site that is easy to reach.

Some households with deep food insecurity never visit a food shelf. Four barriers keep these households away:

- Stigma.
- No knowledge of the program.
- No transport.
- Other obstacles.

This difference matters. It shapes four decisions at The Food Group:

1. How to allocate resources.
2. Where new food shelves are needed.
3. How to argue for a policy.
4. How to measure its own results.

Allocation follows visits today. Visits are a measure of demand, not a measure of need. Allocation therefore tracks demand and not need.

This creates a risk in both directions. An area that is easy to reach generates visits, so it draws allocation. An area of high need with few visits draws little allocation. The visit data gives no signal that the second area exists.

National research supports this concern. Feeding America publishes the Map the Meal Gap study. The study estimates food insecurity for each county. These estimates do not always match the counties with the most food shelf visits.

The Urban Institute runs the Well-Being and Basic Needs Survey. The survey found that approximately 27 percent of working-age adults are food insecure. It found that approximately 17 percent of working-age adults use charitable food. It also found that more than one third of the people who use charitable food are food secure. The gap runs in both directions.

Your task is to build a more complete picture of need and demand. Use the operational data from The Food Group. Also use public data for the economy, for the population, and for food security.

Two measures diverge when they do not agree. Show where need and demand agree. Show where they diverge. Explain the cause.

Datasets from The Food Group

Food shelf activity. The data reports three metrics for each food shelf location and each month:

1. Visits. One visit is one trip by one household. A visit is not a unique person.
2. Pounds distributed. This is the weight of food that leaves the location.
3. Individuals served. The data splits this count into children, seniors, and adults.

The three metrics do not always move together. Treat each metric as a separate signal. Report all three unless a question states otherwise. The data covers 2022 and later years.

Call center logs. Each record is one call. Each record gives the city of the caller, the county of the caller, the reason for the call, and the destination of the referral. A destination is a named food shelf, SNAP enrollment, or another resource. SNAP is the Supplemental Nutrition Assistance Program.

MarketBucks participation. The data gives participation records from the MarketBucks program. The records are grouped by location and by time period.

Organization records. This is the master record for all partner organizations. It covers food banks, food shelves, and other agencies. Each record gives location information and status fields.

Public data sources

Use Feeding America Map the Meal Gap. This source estimates food insecurity for each county. The 2025 edition uses 2023 data. The 2026 edition uses 2024 data. Questions 1 and 6 depend on this source.

Find the other public data yourself. Federal agencies and state agencies publish relevant series. The subjects include prices, income, poverty, population, employment, program enrollment, and access to food retail. The search itself is part of the challenge.

Document each source that you select. For each source, state four things:

1. The publisher and the exact series.
2. The geographic grain, such as the county, the tract, or the region.
3. The time grain, such as monthly, bimonthly, or annual.
4. The reason you chose this source over an alternative.

Divisions and scope

The challenge has three divisions. Each division answers a fixed set of questions. The expectation is that you answer every question in your set. Judges will evaluate your outcomes only on the questions within your set.

Question	Novice	Undergraduate	Graduate
1. Need estimates against observed activity	Yes	Yes	Yes
2. Calls as a signal	Yes	No	No
3. Farmers markets and food shelves	Yes	No	No

4. Food prices and food shelf activity	No	Yes	Yes
5. SNAP enrollment and food shelf use	No	Yes	Yes
6. A divergence measure for need and demand	No	Yes	Yes
7. Forecast August activity	No	Yes	Yes
8. Low-income, low-access areas and the reach of the network	No	No	Yes
Total questions	**3**	**5**	**6**

Four rules set the scope:

1. Every division answers Question 1. It is the foundation for the questions below it.
2. Questions 2 and 3 will be evaluated in the novice division only.
3. Questions 4 through 7 will be evaluated in the undergraduate division and the graduate division.
4. Question 8 will be evaluated in the graduate division only.

The table above determines what outcomes will be judged for each division. It does not limit which data you use. Every dataset is available to every division. An undergraduate team or a graduate team may consider Question 2 or Question 3 and carry their findings into another question.

Questions 6 and 7 each carry a graduate addendum. Graduate teams answer both addenda. Undergraduate teams answer neither.

Question 1: Compare need estimates to observed activity

The ask

Feeding America provides estimates for the number of food-insecure people in each Minnesota county. The Food Group counts the activity at Minnesota food shelves. Do these two sets of numbers tell the same story for the three food shelf metrics (e.g. Visits, Pounds distributed, and Individuals served)? Which metrics tend to have stronger/weaker correlations with the Feeding American estimates? Identify the Minnesota counties where these correlations tend to break down. Using publicly available data, investigate various county characteristics (e.g. poverty rate, the rural share, distance to nearest food shelf, SNAP enrollment, etc) that may help explain why there is a mismatch between the Feeding America estimates and the three food shelf metrics.

Notes:

- Feeding America estimates are found in the Map the Meal Gap dataset
- Use food shelf data from The Food Group for 2023 and 2024

Question 2: Are calls a signal for visits?

The ask

The call center at The Food Group connects people to food resources. If call volume about food assistance goes up in an area, does food shelf activity, with respect to the three food shelf metrics, in that area follow? Do these two signals depend on each other? Are there certain counties that The Food Group should expect more or less types of calls than what were actually received? Identify locations where the calls rise, but the food shelf metrics do not. Investigate potential barriers to food shelf access for these locations.

Notes:

- You should define what constitutes a food-related call.
- Choose a geographic grain and a time grain that the call data can support.

Question 3: Farmers markets and food shelves

The ask

The Minnesota Market Bucks program helps Supplemental Nutrition Assistance Program (SNAP) and SUN Bucks Summer Nutrition Program customers stretch their dollars at farmers markets, making healthy food more affordable. Does access to a farmers market change how much a household uses a food shelf in regards to the three food shelf metrics? Do the two programs serve the same people for the same reasons? Or do they serve different people for different reasons? Do certain counties leverage Market Bucks better than others?

Notes:

- You should limit the analysis to the geographies and the time periods for which these data sources overlap.
- One issue that should be dealt with here is the timing of when farmers markets are available. For example, a farmers market may only be open for the summer months or a new farmers market may just pop up in an area. How do these type of changes affect a local food shelf?

Question 4: Food prices and food shelf activity

The ask

Do more people arrive at food shelves when grocery prices rise? Does the response happen at once, or is there a delay? Does the response appear in all three of the food shelf metrics? Are there other cyclical patterns that prices do not explain? Understanding the timing enables The Food Group to communicate expected changes in demand to food shelves and expected need to donors.

Question 5: SNAP enrollment and food shelf use

The ask

SNAP gives a household money to buy food. A food shelf gives a household food directly. Do the two move together, or does one replace the other? The SNAP benefits may change from time-to-time, what happens to food shelf activity when the SNAP benefits change? Does the relationship between SNAP benefits and food shelf activity differ across locations? Understanding the nature of this relationship helps The Food Group highlight the importance of SNAP to lawmakers and in turn helps them make better informed decisions.

Notes:

- You should find a public SNAP source that can be joined to the food shelf data, with the source, the grain, and the publication lag stated.

Question 6: Build a divergence measure for need and demand

The ask

Food shelf activity shows where people arrive. It does not show where people need help and do not get it. Create a new metric or measure that highlights the areas where need is likely larger than the activity data shows. Compare and contrast this measure of need against the three food shelf metrics. Are there locations where your new metric of need and the three food shelf metrics diverge from each other? Akin to Question 1, investigate various county characteristics that might help explain why there is disagreement between your new measure of need and food shelf metrics.

Notes:

- Your new metric or measure of need should consider more than just demand.
- A sufficient and complete measure of need will likely require the use of other publicly available data.

Graduate addendum

A falsifiable definition is a definition that evidence can prove wrong.

1. A falsifiable definition of need, stated plainly.
2. An explanation of why your new metric captures need rather than demand.
3. The evidence that would make you reject the definition.
4. The evidence that would make you revise the measure.

Question 7: Forecast August activity

The ask

The food shelf data you receive ends in July. The prediction component of this year's competition requires that you predict the August result for each of the three reported metrics: visits, pounds distributed, and individuals served for MN Counties.

August has already happened. You do not have its data, and you must not go looking for it. Use of any August data from any source will disqualify your team from this year's competition.

Graduate addendum

1. A separate prediction for children served, seniors served, and adults served.

Question 8: Low-income, low-access areas and the reach of the network

The ask

Some areas have a large number of households with low income and may have poor access to a grocery store. Does the food shelf network sufficiently reach these areas? Does the food shelf network fill the retail gap, or does it repeat the same pattern as the retail market? Akin to question 1, investigate any differences across various locations. Discuss any limitations of your outcomes from this question.

Notes:

- A low-income, low-access area is a census tract where household incomes are low and the distance to a grocery store is large. USDA shortens this to LILA. Older sources call the same areas food deserts.
- You will need to find a public data source that identifies low-income, low-access areas, with its definition and the vintage of its underlying data stated.